

1. Addressing modes

Given

Register	Value
PC	4200H
R1	1500H
R2	2500H
IX	0800H
SP	3000H

Memory

Address	Data
1500H	45H
1502H	3200H
2500H	18H
2800H	3400H
3000H	3600H
3200H	55H
3400H	66H
3600H	77H
4200H	200H

Instruction operand = 200H

a.) Find Effective Address (EA) and operand

1.) Immediate Addressing

EA = not required

Operand = 200H

The value is directly inside the instruction

2. Direct Addressing

EA = operand

EA = 200H

Data is Stored at memory

location 200H

Operand = Not Given

3. Indirect Addressing

operand = 200H

operand field Contains an address which stores another address

EA = Cannot be determined

4. Register Addressing

Data is directly inside the register.

R₁ is used

R₁ = 1500H

EA = Register R₁

Operand = 1500H

5. Register Indirect Addressing

Register Contains the memory address

R₁ = 1500H

Memory [1500H] = 45H

EA = 1500H

operand = 45H

6. Indexed Addressing

EA = IX + operand

IX = D800H

operand = 200H

EA = DA00H

operand = not Given

7. Relative Addressing

EA = PC + operand

PC = 4200H

operand = 0200H

EA = 4400H

operand = not Given

8. Stack Addressing

Top of Stack is Pointed by SP

SP = 3000H

Memory [300H]

= 3600H

Memory [3600H]

= 77H

EA = 3000H

Operand = 77H

b) which addressing mode requires maximum memory access?

Indirect Addressing

Reason: Because it accessed memory two times

Example

Instruction

↓
memory (gets address)

↓
memory again (gets data)

So it takes more time.

c. Arrange in increase order of execution speed.
Fastest → slowest

Register

↓

Immediate

↓

Register Indirect

↓

Direct

↓

Relative

↓

Indexed

↓

Stack

↓

Indirect

Reason: Register is fastest because CPU already has the data

Indirect is slowest because it accesses memory multiple times

- 2 A Computer executes 900 million instruction with an average CPI of 2.8. Initially, the processor runs at 2.4 GHz. Later the clock frequency is upgraded to 3.6 GHz without changing CPI.

GIVEN

Instruction Count = 900 million

CPI = 2.8

Clock Frequency = 2.4 GHz

FORMULA:

$$\text{CPU Time} = \frac{(\text{Instruction Count} \times \text{CPI})}{\text{Clock Frequency}}$$

i) Original Execution Time

$$\begin{aligned}\text{Clock Cycles} &= 900 \times 10^6 \times 2.8 \\ &= 2520 \times 10^6\end{aligned}$$

$$\text{Clock Frequency} = 2.4 \times 10^9$$

$$\text{Execution Time} = \frac{2520 \times 10^6}{2.4 \times 10^9}$$

$$= 1.05 \text{ Seconds}$$

ii) New Execution Time

$$\text{Frequency} = 3.6 \text{ GHz}$$

$$\frac{2520 \times 10^6}{3.6 \times 10^9} = 0.70 \text{ Seconds}$$

iii) Speedup

Formula:

$$\frac{\text{Old Time}}{\text{New Time}} = \frac{1.05}{0.70} = 1.5$$

iv) CPI increased by 10%.

$$\text{New CPI} = 2.8 \times 1.10 = 3.08$$

$$\text{Clock cycles} = 900 \times 10^6 \times 3.08 = 2772 \times 10^6$$

$$\text{Execution Time} = 2772 \times 10^6 / 3.6 \times 10^9 = 0.77 \text{ Seconds}$$

v) Is increasing clock frequency always beneficial?

No.

Although frequency increased, CPI also increased

Execution time became
 $0.70 \text{ s} \rightarrow 0.77 \text{ s}$

So higher frequency does not always mean better performance.

3.

Instruction Mix

Instruction	Count	CPI
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Arithmetic	5×10^6	2
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Memory	3×10^6	5
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Branch	1×10^6	4
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Floating	2×10^6	8
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Clock Frequency = 2.5 GHz

i) Total Clock Cycles

Arithmetic $5 \times 2 = 10 \text{ million}$

Memory $3 \times 5 = 15 \text{ million}$

Branch $1 \times 4 = 4 \text{ million}$

Floating
 $2 \times 8 = 16$ million

$$\text{Total} = 10 + 15 + 4 + 16 = 45 \text{ million cycle}$$
$$= 45 \times 10^6 \text{ cycles.}$$

ii) CPU Time

$$45 \times 10^6 / 2.5 \times 10^9$$
$$= 0.018 \text{ seconds}$$

iii) Average CPI

Total instruction
 $5 + 3 + 1 + 2 = 11$ million

$$\text{Average CPI} = 45 / 11 = 4.09$$

iv) Memory CPI reduced to 3
New memory cycles
 $3 \times 3 = 9$

$$\text{Total } 10 + 9 + 4 + 16 = 39 \text{ million}$$

$$\text{CPU Time } 39 \times 10^6 / 2.5 \times 10^9 = 0.0156 \text{ s}$$

v) Percentage Improvement

$$(0.018 - 0.0156) / 0.018 \times 100 = 13.33\%$$

4 Processor Comparison

Formula:

$$\text{Execution Time} = (\text{Instruction Count} \times \text{CPI}) / \text{clock Frequency}$$

Processor A

$$500 \times 2.5 = 1250$$

$$1250 / 4000 = 0.3125 \text{ s}$$

Processor B $550 \times 1.8 = 990$

$$= 990 / 3000 = 0.33 \text{ s}$$

Processor C

$$430 \times 3.1 = 1333$$

$$1333 / 5000 = 0.2666s$$

1. Execution Time

Processor	Time
A	0.3125s
B	0.33s
C	0.2666s

2. Ranking

Fastest
C
↓
A
↓
B

3. Speedup

$$0.33 / 0.2666 = 1.24$$

4. B reduces instruction Count by 10%.

New Instruction $550 \rightarrow 495$ million

$$\text{Execution Time} = 495 \times 1.8 / 3000 \\ = 0.297s$$

5. Best Processor.

Processor C

Reason:

- * Highest clock frequency
- * Lowest execution time
- * Best Overall Performance